

ALI SHAHIDI

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SUMMARY

Multidisciplinary software developer specializing in Unreal Engine, Unity, C++, and XR technologies. Experienced in building reusable Blueprints, optimizing game systems, and extending engine functionality through custom plugins and tools. Proven track record of delivering high-performance, interactive applications in VR, AR, and real-time 3D environments. Passionate about experimentation, performance tuning, and pushing the limits of game engines to craft immersive, technically robust experiences.

PROFESSIONAL EXPERIENCE

Software Developer *September 2024 - November 2025*
ETHEREAL Research Group *Calgary, AB*

- Collaborated in the design of an interactive VR training environment for autistic users, focusing on accessibility, interaction flow, and ethical design.
- Engineered backend systems to synchronize GPT-driven dialogue, user events, and sensory feedback in real time across multimodal inputs.
- Incorporated and implemented AI-driven real-time behavioral analysis, boosting video analysis speed by 90% and accuracy by 83% with local algorithms.
- Optimized performance by reducing third-party API costs by 98% through adaptive frame rate control and delta-based body language analysis.

Software Developer - Protégé *September 2024 - March 2025*
ELIXR & Illumia *Calgary, AB, Remote*

- Contributed to 5 cross-functional projects, delivering a scalable solution for VR/AR experiences. focusing on areas such as animation, lip-syncing, AI integration, performance tuning, real-time audio streaming, real-time Speech-to-Text processing, and interactive WebXR, Mobile, VR/AR, and PC experiences.
- Led and mentored a new intern by explaining code architecture, guiding implementation steps, and coordinating tasks through weekly sync meetings.

Software Developer - Internship *February 2024 - August 2024*
Telus *Calgary, AB*

- Collaborated with Telus Communications to develop a cross-platform AR application for iOS and Meta Quest 2.
- Engineered multiplayer functionality with NetCode and spatial anchor integration for AR app on iOS/Meta Quest 2.

Graduate Research & Teaching Assistant - Software Engineer *January 2021 - April 2024*
University of Calgary *Calgary, AB*

- Led a study on assistive XR technology in collaboration with psychologists and professionals.
- Provided Unity support during a 2-day IEEE Tech-Hackathon (SIGHT).
- Delivered courses on Software Engineering, OOP, Scalable Data Analysis, and Programming Fundamentals.

EDUCATION

University of Calgary *April 2024*
Master of Software Engineering (GPA: 4/4.00) *Calgary, AB*

- Received full departmental graduate funding and international graduate tuition award.
- Nominated for the Best Graduate Project at the Distinguished Master's Thesis Award in STEM Disciplines.

University of Tehran *July 2020*
B.Sc. in Computer Engineering (GPA: 17.24/20.00) *Tehran, Iran*

- Received the best undergraduate project, and two Faculty of Engineering Fellowship Awards.

PROJECTS

ETHEREAL Research Group - Coffee Shop Simulation | Unreal Engine, C++, CC5, Chat GPT, API, VR

- Built an immersive VR coffee shop simulator to help autistic users practice communication and decision-making in a realistic social setting.
- Integrated GPT, ElevenLabs, and CC5 avatar creation for real-time avatar narration, adaptive dialogue, and event-driven responses within the virtual space.
- Designed and implemented reusable Unreal Engine Blueprints and Editor Utility Widgets to automate level generation and scene organization.
- Extended Unreal Engine 5 features using C++ and Python, developing modular plugins for UI, gameplay events, and asset management.
- Developed a procedural VR training simulation leveraging Unreal Engine 5's PCG system and Control Rig to dynamically adjust environments to user performance.

ETHEREAL Research Group - Multi Modal Behaviour Analysis | Unity, AI, VR/AR/Tablet, Python

- Built a cross-platform real-time video analysis app for tablets and Quest 3, enabling multimodal (text, video, voice, image) input processing.
- Designed a server-based AI pipeline to offload VR computation and support large-file chunking for efficient, reliable data transmission.

ELIXR & Illumia | Unreal Engine, Unity, C#, C++, Network, Optimization, VR, Azure

- Built AI-driven animation system using third-party APIs to reflect user emotions.
- Improved performance via Profiler, GPU Instancing, Static Batching, Occlusion Culling, and Object Pooling.
- Real-time data streaming enabled with WebSocket integration.
- Refactored and scaled codebases for projects including flight simulators, training tools, and virtual assistants.

Telus | Unity, Unity 3D, NetCode, Azure

- Led the development of a multiplayer Augmented Reality application with Telus Communications, featuring NetCode, mapping, mesh rendering, animation, and interactions.

HoloType | Unity, C#, MRTK, Scrum

- Designed and led development of *HoloType*, a modular 3D AR educational app for HoloLens 2.
- Integrated photon-based multiplayer, improving the user experience by 25%.
- Conducted iterative user studies, enhancing usability and reducing caregiver support needs by 35%.
- Added customization features, increasing user satisfaction by 20%.
- Project secured additional funding and led to three published papers.

Complete C# Unity Game Developer 3D Course | Unity, C#

- Engineered realistic terrains, timeline animations, and applied post-processing effects in Unity; optimized pathfinding algorithms and grid systems to enhance gameplay experience and performance.
- Leveraged Unity's powerful features, including particle system and Cinemachine, to create visual effects and captivating game sequences.

SKILLS

Languages: Unreal Engine 5 (Blueprints, Control Rig, PCG, EUW, Plugins) C++, C#, Unity, Unity 3D, Python

Frameworks: Express.js, React.js

Databases: PostgreSQL, MySQL, SQL

Tools: Git, ClickUp, Docker, ARKit, ChatGPT, Cursor, Scrum, Jira, Confluence, LaTeX

CERTIFICATIONS

Programming Design Patterns For Unity: Write Better Code | Udemy

July 2024

The Beginner's Guide to Animation in Unity | Udemy

June 2024

The Complete Web Development Bootcamp | Udemy

January 2024

Unity C# Mobile Game Development: Make 3 Games From Scratch | Udemy

October 2023

Complete C# Unity Game Developer 3D | Udemy

September 2023